

Generation Interconnection System Impact Study Report

For Queue Project Q20-1587

Mickens 138 kV

June 28, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1587. The Interconnection Customer proposes a 299.7 MW solar plus storage facility interconnecting at the Mickens 138 kV substation in the Virginia area. The request was received on October 06, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 451 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Skags Island Jct 2 Substation / Mitchell Dam Substation 345.0 kV, the Westcoast Cellufibre / Leavitt 138.0 kV Cir. 1 loads to 103.3% of its 384 MVA rating. Project DFAX: 7.7%.

For the loss of Mitchell Dam / Skaggs Island Junction 2 345.0 kV, the Westcoast Cellufibre / Leavitt 138.0 kV loads to 103.3% of its 384 MVA rating. Project DFAX: 7.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Westcoast Cellufibre / Leavitt 138.0 kV #1	Reconductor line	\$450,000	3% (\$14,000)
Leavitt / Westcoast Cellufibre 138.0 kV Cir. 2	Reconductor line	\$645,000	3% (\$20,000)
	Total allocated to project		\$34,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.